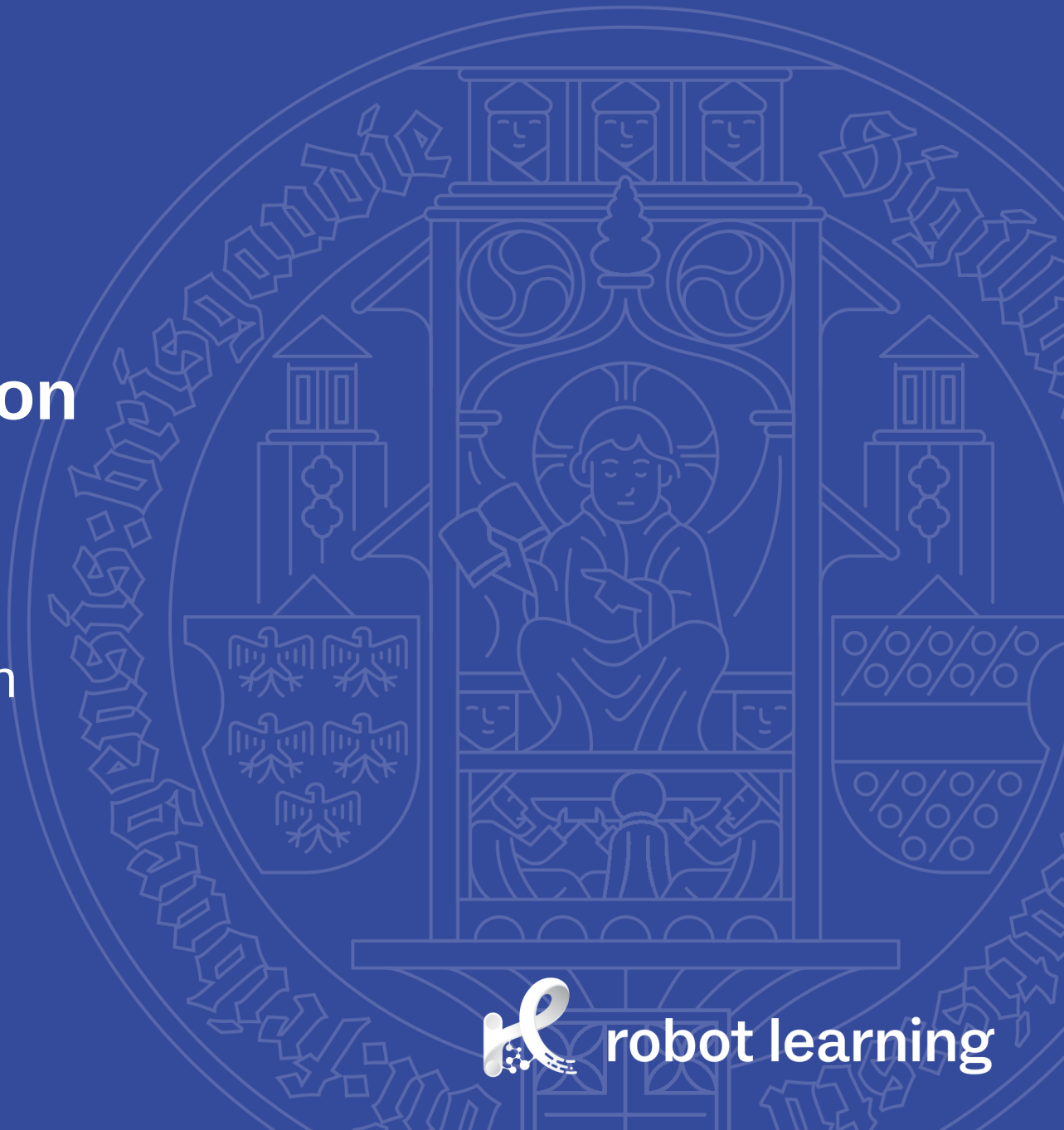
Label-Free Class Incremental 3D Semantic Scene Completion

Tushar Samal | MSc. Project Presentation

Supervisor: Prof. Dr. Abhinav Valada
Advisors: Nicolas Schischka and Julia Hindel

19 October 2026



robot learning

Agenda
Outline

1. Introduction & Motivation
2. Related Work
3. Methodology
4. Experimental Results & Conclusion

Part 1

Introduction & Motivation

Semantic scene completion and learning classes over time



Introduction

3D semantic scene completion

INPUT



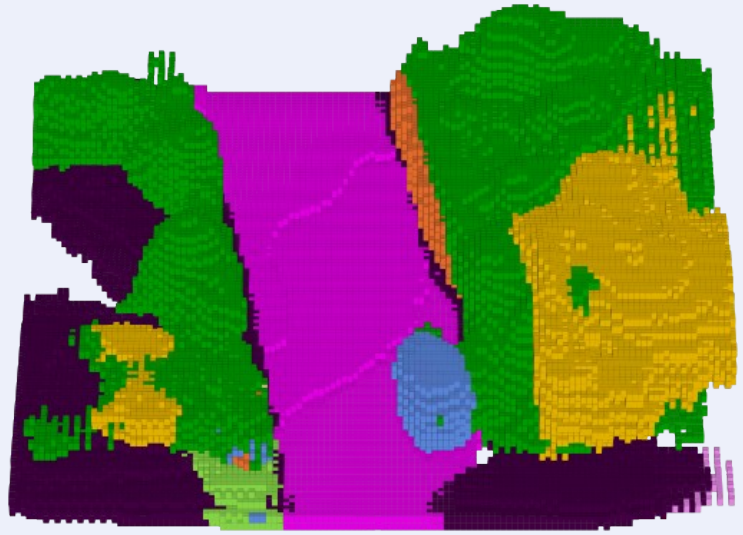
One RGB image from a single front camera

Semantic scene completion

DISC
camera-only model

a semantic class for every voxel, visible or not

OUTPUT



A class for every voxel of the 51.2 m × 51.2 m × 6.4 m volume

Introduction

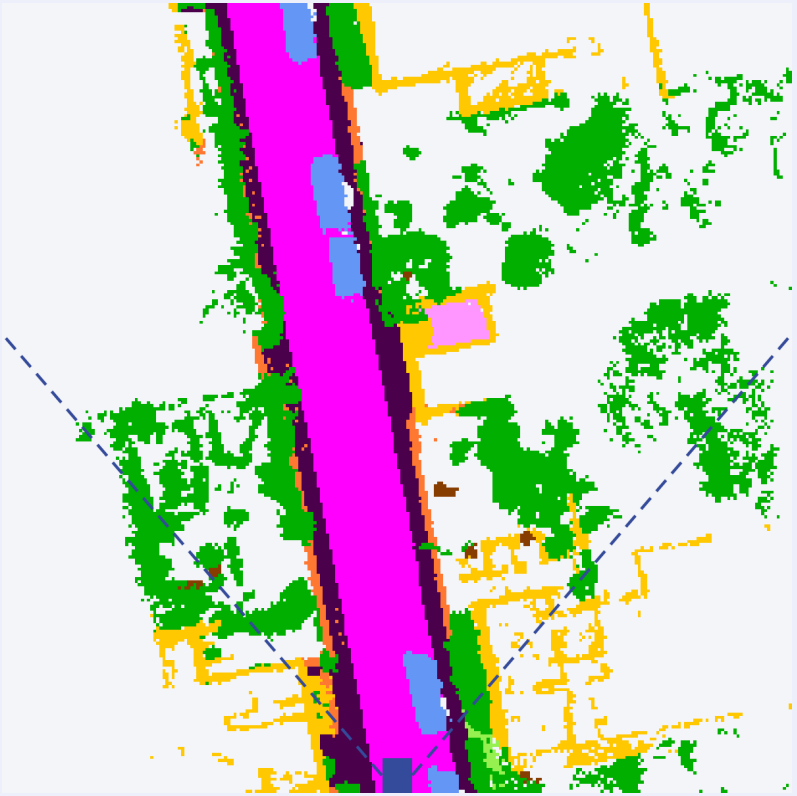
What the 3D output contains



One RGB image from a single front camera

Ground-truth semantic occupancy, seen from above

51.2 m ahead · 51.2 m wide · 6.4 m high · 256 × 256 × 32 voxels of 0.2 m



Motivation

Why add classes step by step?

The real world

- **New classes** appear over time
- **Retraining** from scratch is costly

The challenge: forgetting

- **Learning new classes** erodes old ones
- **Old labels** are not available

Our setting: disjoint class-incremental learning

Motivation

Continual learning in 5 steps: classes arrive one group at a time



Step 0
seq 01, 10

road | **sidewalk** | **vegetation** | **terrain**

4 classes known



Step 1
seq 03, 04

4 old classes
no labels

fence | **trunk** | **pole**

7 classes known



Step 2
seq 05, 09

7 old classes
no labels

car | **person** | **traffic-sign**

10 classes known



Step 3
seq 06, 07

10 old classes
no labels

truck | **parking** | **building**

13 classes known



Step 4
seq 00, 02

13 old classes
no labels

bicycle | **motorcycle** | **bicyclist** | **motorcyclist**

17 classes known

old classes: labels not available

new classes: the only labels at this step

The previous model is the only memory of the old classes

Motivation

Research questions

RQ1 New classes

Can 2D masks from an open vocabulary model replace manually annotated 3D labels?

RQ2 Replay

How much does a replay buffer alleviate catastrophic forgetting?

RQ3 Long-term knowledge retention

How much does the performance differ from three to five steps?

Part 2

Related Work

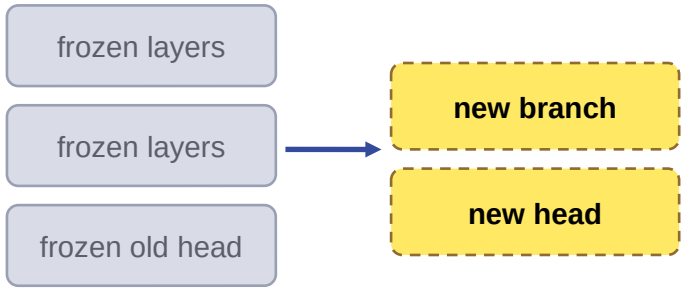
Knowledge retention in continual learning



Related Work

Three ways to retain knowledge in continual learning

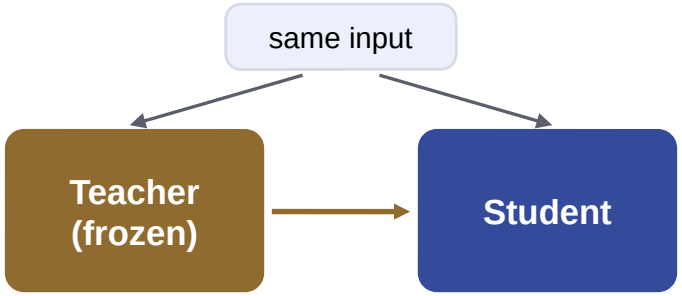
Network expansion



Add new parameters, keep the old weights frozen.

ECLIPSE [1]

Knowledge distillation

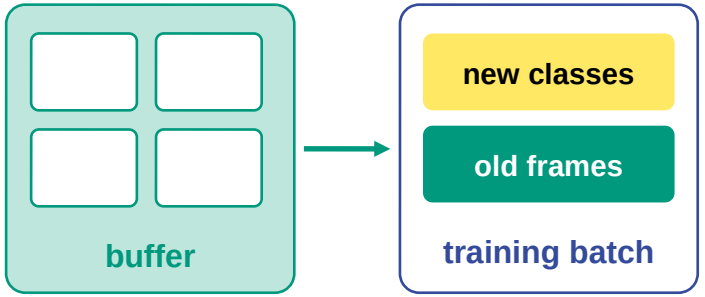


A frozen copy of the old model teaches the new one.

CIT [2]

Incrementer [3]

Replay



Train again on stored old samples.

Replay Master [4]

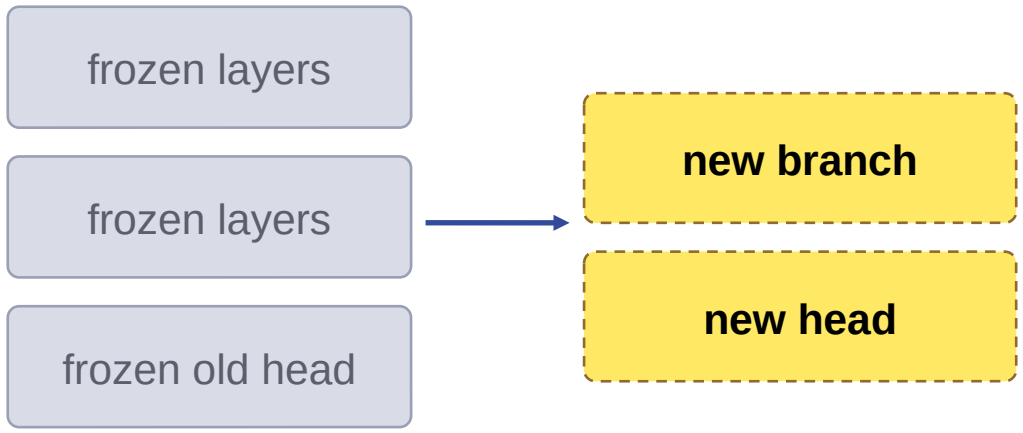
MiB-AugM [5]

[1] Kim et al., "ECLIPSE: Efficient Continual Learning in Panoptic Segmentation with Visual Prompt Tuning", CVPR 2024
[2] Ge et al., "CIT: Rethinking Class-incremental Semantic Segmentation with a Class Independent Transformation", arXiv 2024
[3] Shang et al., "Incrementer: Transformer for Class-Incremental Semantic Segmentation with Knowledge Distillation Focusing on Old Class", CVPR 2023
[4] Zhu et al., "Replay Master: Automatic Sample Selection and Effective Memory Utilization for Continual Semantic Segmentation", IEEE TPAMI 2025
[5] Kwak et al., "Towards Realistic Incremental Scenario in Class Incremental Semantic Segmentation", CoLLAs 2024

Related Work

Network expansion

Network expansion



In our project

Only the output head grows: one node per new class, old nodes are copied from the previous model.

How it works

1. Freeze the layers that learned the old classes
2. Add new capacity: a branch, a head or prompts
3. Train only the new part on the new data

Pro

Frozen parts cannot forget

Con

The model grows with every step

Key papers

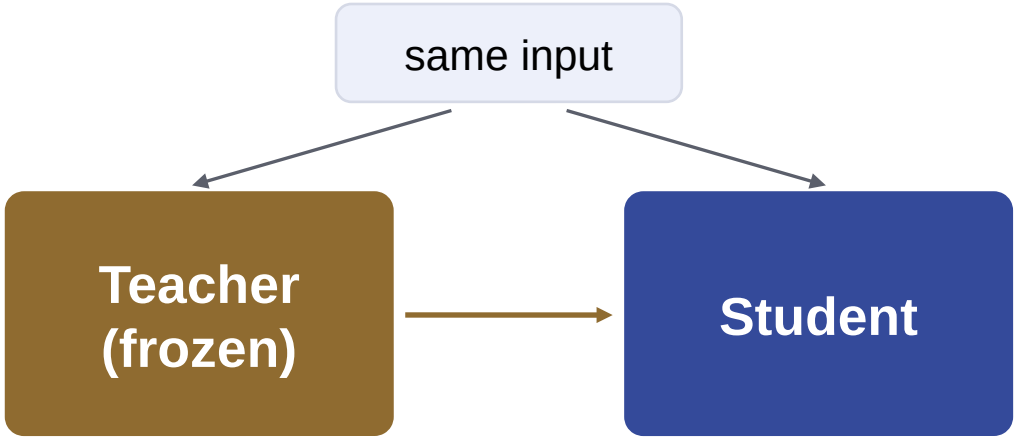
ECLIPSE [1]

[1] Kim et al., "ECLIPSE: Efficient Continual Learning in Panoptic Segmentation with Visual Prompt Tuning", CVPR 2024

Related Work

Knowledge distillation

Knowledge distillation



In our project

The previous model gives pseudo-labels for old classes, only where it is confident (≥ 0.8).

How it works

1. The previous model becomes a frozen teacher
2. Teacher and student see the same input
3. The student must match the teacher on old classes

Pro

No old data has to be stored

Con

Teacher errors are inherited

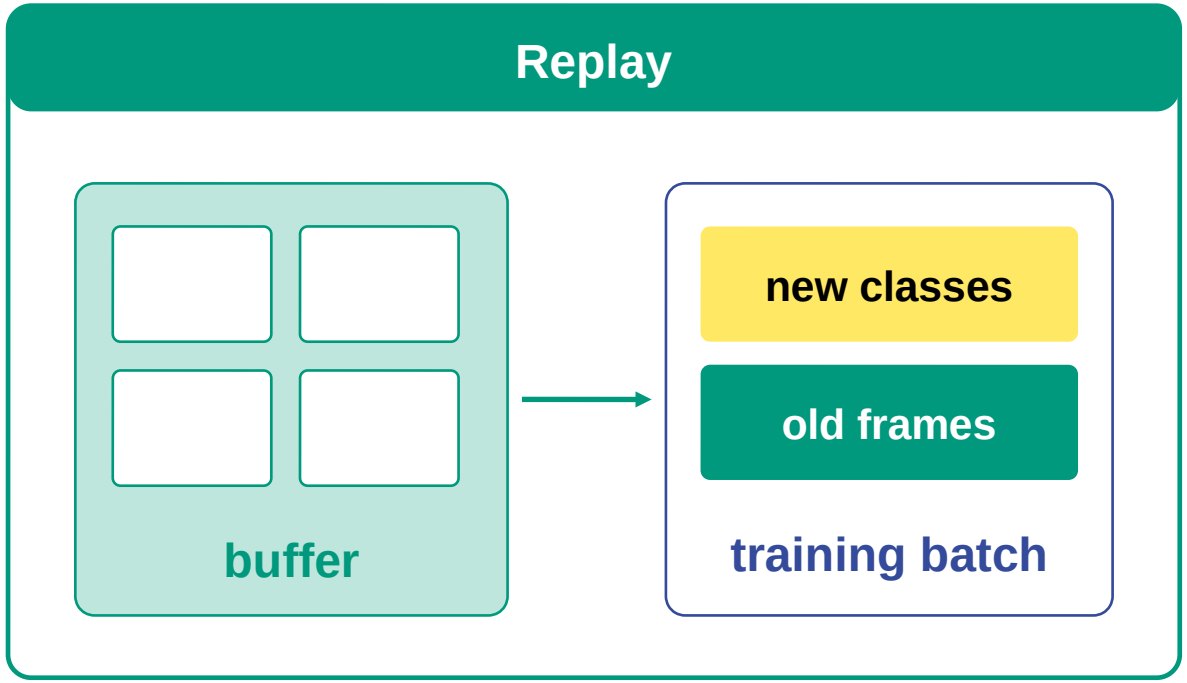
Key papers

CIT [1] | Incrementer [2]

[1] Ge et al., "CIT: Rethinking Class-incremental Semantic Segmentation with a Class Independent Transformation", arXiv 2024
[2] Shang et al., "Incrementer: Transformer for Class-Incremental Semantic Segmentation with Knowledge Distillation Focusing on Old Class", CVPR 2023

Related Work

Replay



In our project

A buffer of 49 real frames with real old-class labels, selected class-aware and mixed into every batch.

How it works

1. Keep a few old samples
2. Mix them into every training batch of the new step
3. The model keeps seeing the old classes

Pro

Real labels, simple to add

Con

Needs memory; which samples to keep matters

Key papers

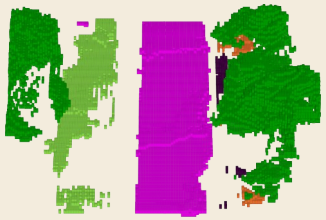
- Replay Master [1]
- MiB-AugM [2]

[1] Zhu et al., "Replay Master: Automatic Sample Selection and Effective Memory Utilization for Continual Semantic Segmentation", IEEE TPAMI 2025
[2] Kwak et al., "Towards Realistic Incremental Scenario in Class Incremental Semantic Segmentation", CoLLAs 2024

Related Work

Why we use a replay buffer

Pseudo-labels



imperfect predictions, but cover every frame

+

Replay frames



real labels, but only 49 frames

→

Old classes kept

each source covers the other's gaps

1

No model change

data loading only

2

Real labels

not teacher predictions

3

Cheap

49 frames in total

4

Complements the previous model

fixes what its pseudo-labels get wrong

Part 3

Methodology

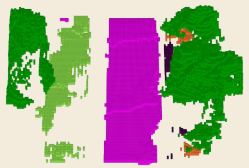
DISC and our continual-learning extension



Methodology

Our class-incremental framework at step t

Old classes: pseudo labels



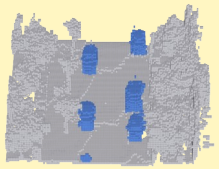
pseudo-labels where the previous model is confident (≥ 0.8)

Old classes: replay buffer



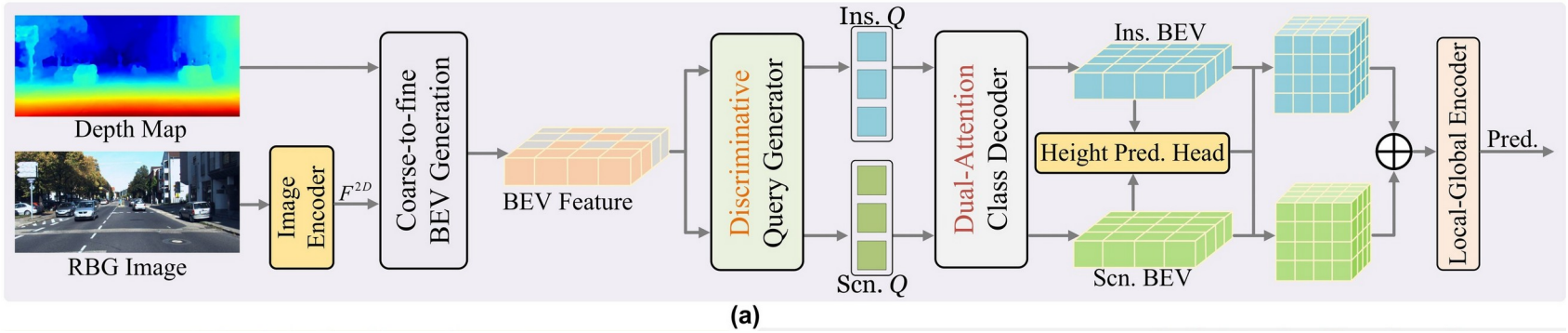
49 stored frames with real old-class labels

New classes: label source



3D GT, 2D GT or SAM3 labels of this step's classes

Student: DISC [1]



output head grows



one node per class

old: copied

new: added

3D losses *or* **2D render CE** Σ **Pseudo-label CE** Σ **Replay CE** → **✓ Training complete Becomes the checkpoint of step t+1**

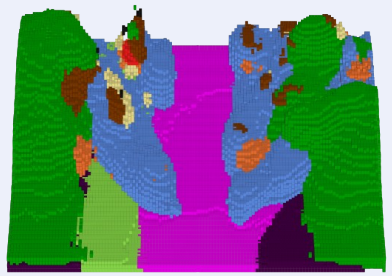
[1] Liu et al., "Disentangling Instance and Scene Contexts for 3D Semantic Scene Completion", ICCV 2025

Methodology

Supervision 1: 3D ground truth for the new classes

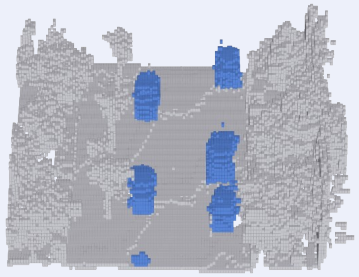
New classes: 3D ground truth

3D prediction of the student



all classes of step 2 are predicted

3D ground-truth label



new classes in colour; grey = not supervised

prediction

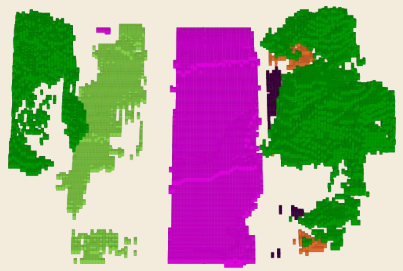
target

3D losses, computed on the new-class voxels

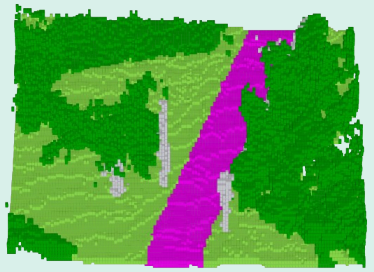
- **Cross-entropy**
- **Semantic scal**
- **Geometric scal**

Old classes: same in every variant

Pseudo-label CE



Replay CE



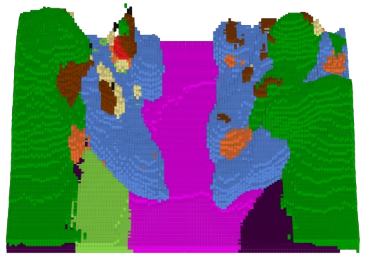
Methodology

Supervision 2: 2D ground truth and the render loss

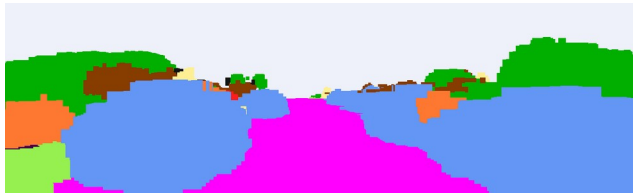
New classes: 2D ground truth

Render loss

3D prediction (student)



prediction: rendered 2D view



target: 2D GT mask (new classes)

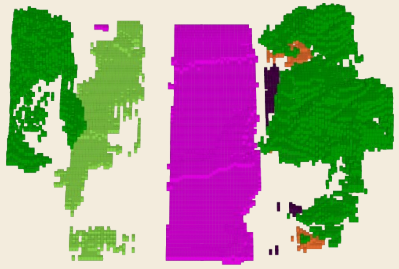


2D render CE
new classes only

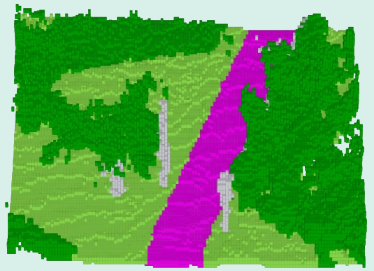
Rendering supervision follows RenderOcc [1]

Old classes: same in every variant

Pseudo-label CE



Replay CE



[1] Pan et al., "RenderOcc: Vision-Centric 3D Occupancy Prediction with 2D Rendering Supervision", ICRA 2024

Methodology

Supervision 3: SAM3 masks from text prompts

New classes: SAM3 masks

camera frame



SAM3 [1]

SAM3 mask (new classes)

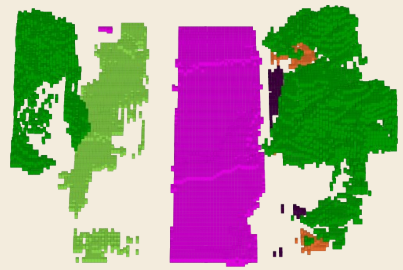


	single prompt	multi-prompt
car	car	car · sedan · van
person	person	person · pedestrian
traffic-sign	traffic-sign	traffic-sign · road-sign

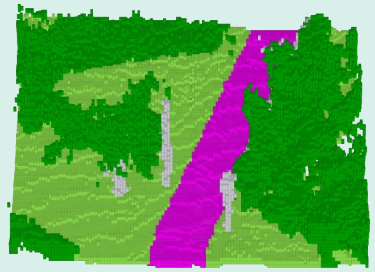
Then the same render loss as for 2D GT

Old classes: same in every variant

Pseudo-label CE



Replay CE

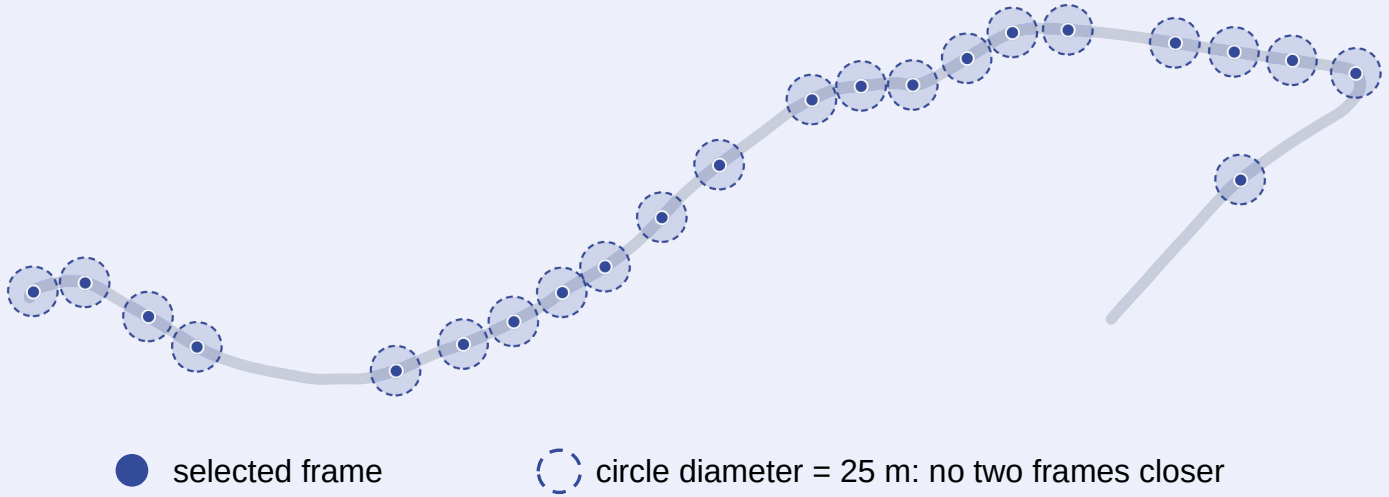


[1] Carion et al., "SAM 3: Segment Anything with Concepts", arXiv 2025

Methodology

Replay buffer: class-aware selection, 25 m apart

Sequence 10: the 22 frames chosen for the step-1 buffer



selected frame circle diameter = 25 m: no two frames closer

Random replay (comparison): same budget of about 49 frames, but frames are drawn by chance

1 Coverage

rare old classes count more

2 Hardness

hard frames first

3 Spread

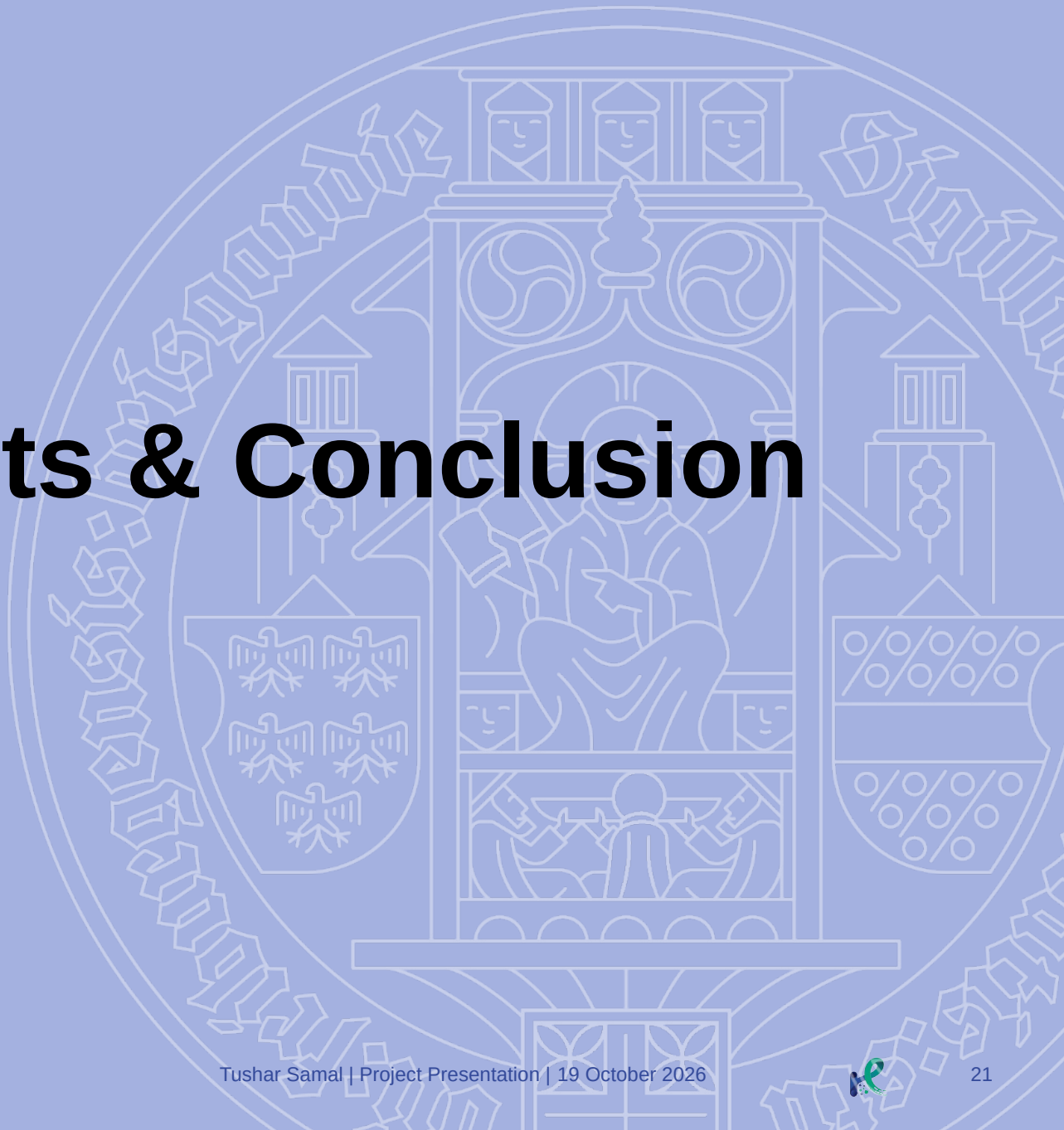
≥ 25 m apart, 49 frames

Buffer frames keep their real labels

Part 4

Experimental Results & Conclusion

Label sources, replay, 3 and 5 steps



Experimental Results

Experimental setup

Protocol

- **Dataset:** SemanticKITTI [1]
- **Validation:** sequence 08
- **Setting:** disjoint class-incremental
- **Metrics:** mIoU, base mIoU, old mIoU, novel mIoU
- **Replay:** 49 frames, class-aware or random

3-step split [2]

Step	Sequences	New classes
0	01, 02, 03	road, parking, sidewalk, other-ground, vegetation, terrain
1	04, 05, 09, 10	building, fence, trunk, pole, traffic-sign
2	00, 06, 07	car, bicycle, motorcycle, truck, other-vehicle, person, bicyclist, motorcyclist

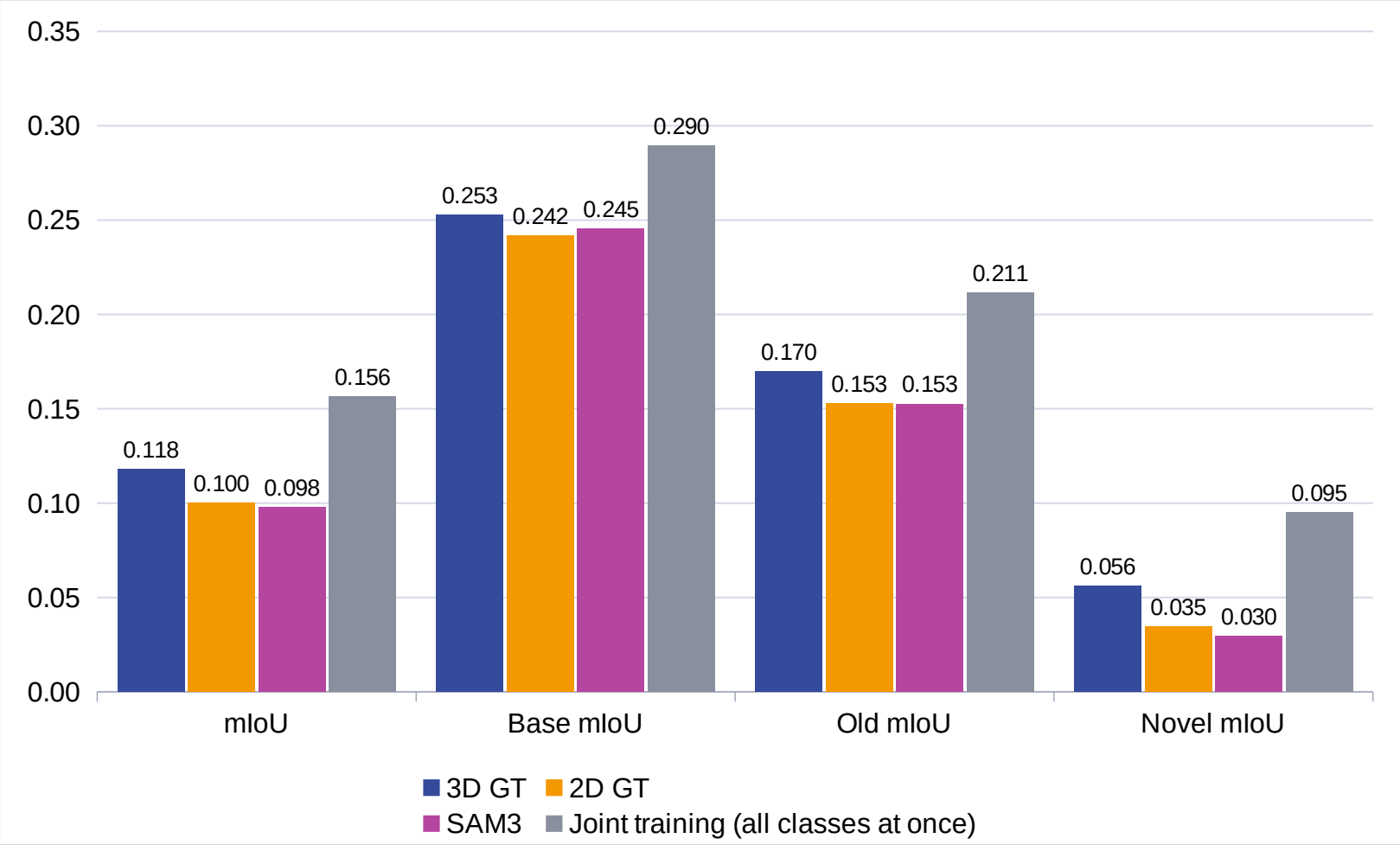
5-step split

Step	Sequences	New classes
0	01, 10	road, sidewalk, vegetation, terrain
1	03, 04	fence, trunk, pole
2	05, 09	car, person, traffic-sign
3	06, 07	truck, parking, building
4	00, 02	bicycle, motorcycle, bicyclist, motorcyclist

[1] Behley et al., "SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences", ICCV 2019
[2] Liu et al., "Disentangling Instance and Scene Contexts for 3D Semantic Scene Completion", ICCV 2025

Experimental Results

3-step, final step: label sources without replay



85%

2D GT vs. 3D GT mIoU (0.1003 vs. 0.1180)

83%

SAM3 vs. 3D GT mIoU (0.0977 vs. 0.1180)

0.156

mIoU of joint training, the upper bound; 3D GT reaches 75% of it

Experimental Results

3-step, final step: single vs. multi-prompt SAM3



0.0977

mIoU with a single prompt, vs. 0.0736 – 0.0973 for the multi-prompt recipes

No gain

using SAM3 for old classes: 0.0736 – 0.0963 vs. 0.0973 with pseudo labels for old classes

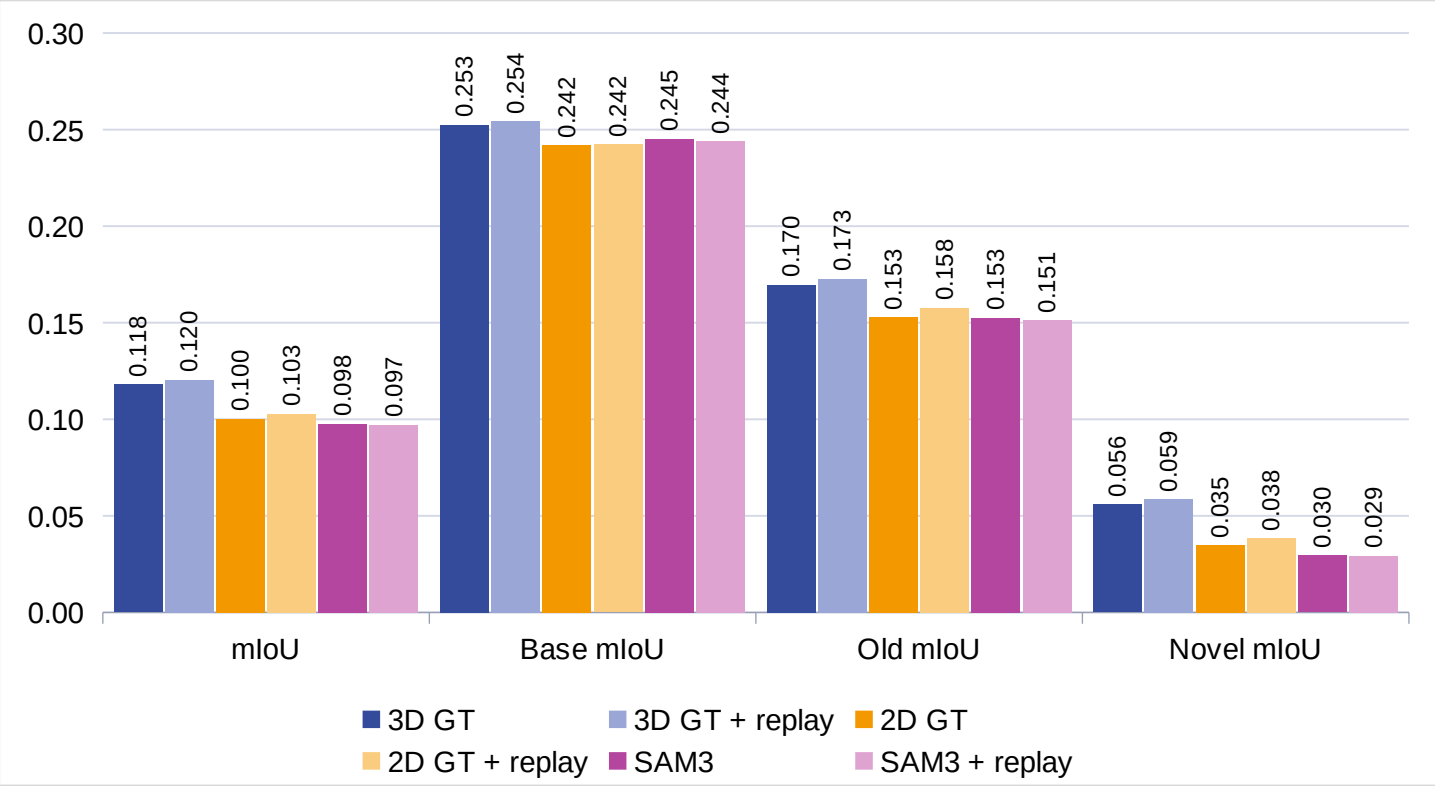
Single prompt only

Later experiments use one prompt per class for the new classes; old classes use pseudo-labels

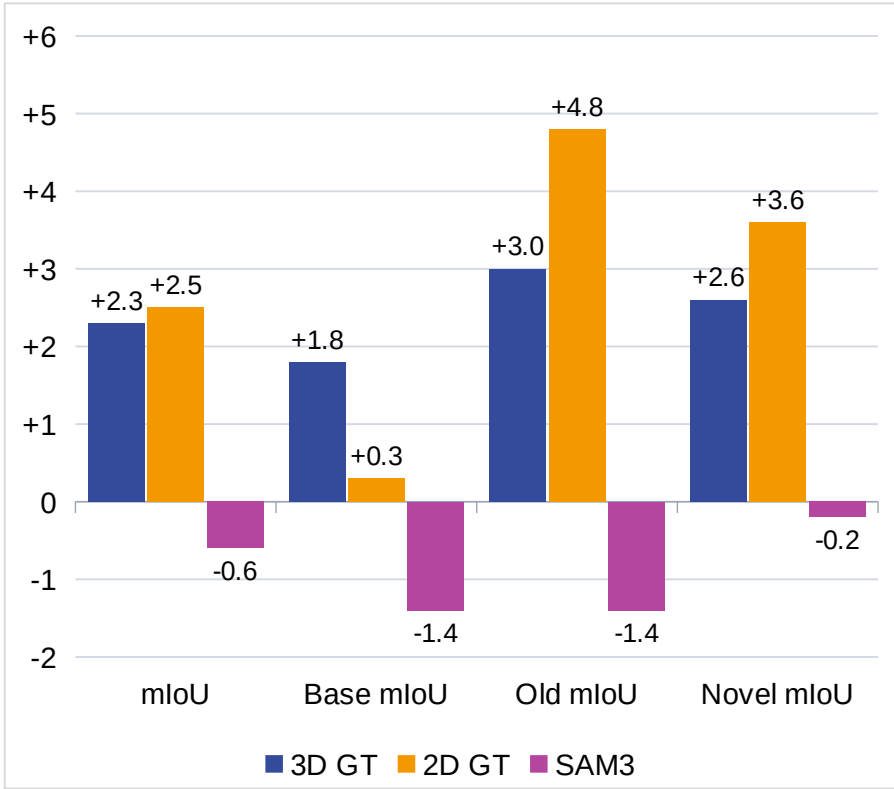
Experimental Results

3-step, final step: what replay adds

Without and with replay (49 frames)



Change from replay (× 10⁻³)

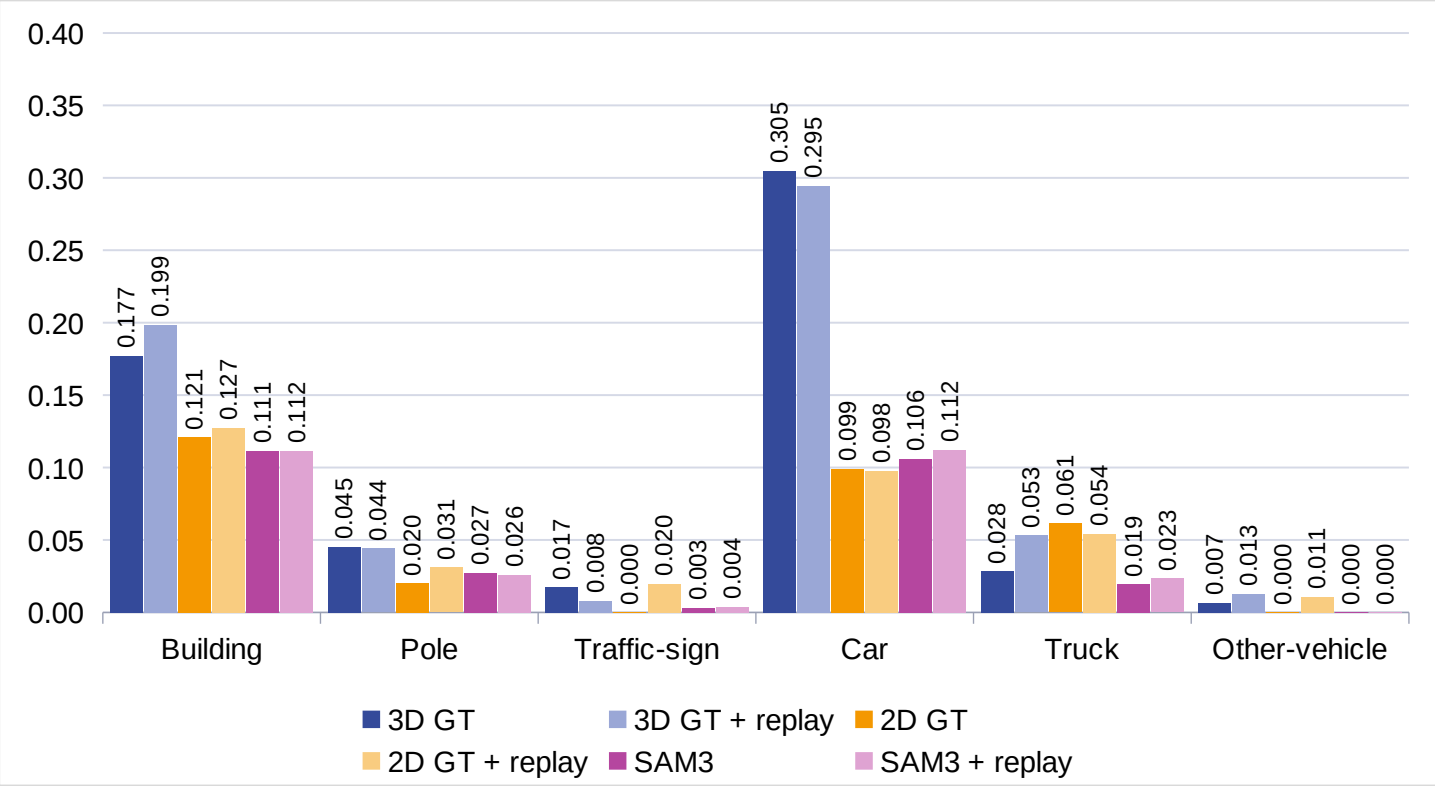


Replay helps 3D GT and 2D GT a little (mIoU +0.0023 / +0.0025, Old mIoU +0.0030 / +0.0048); SAM3 does not benefit (−0.0006).

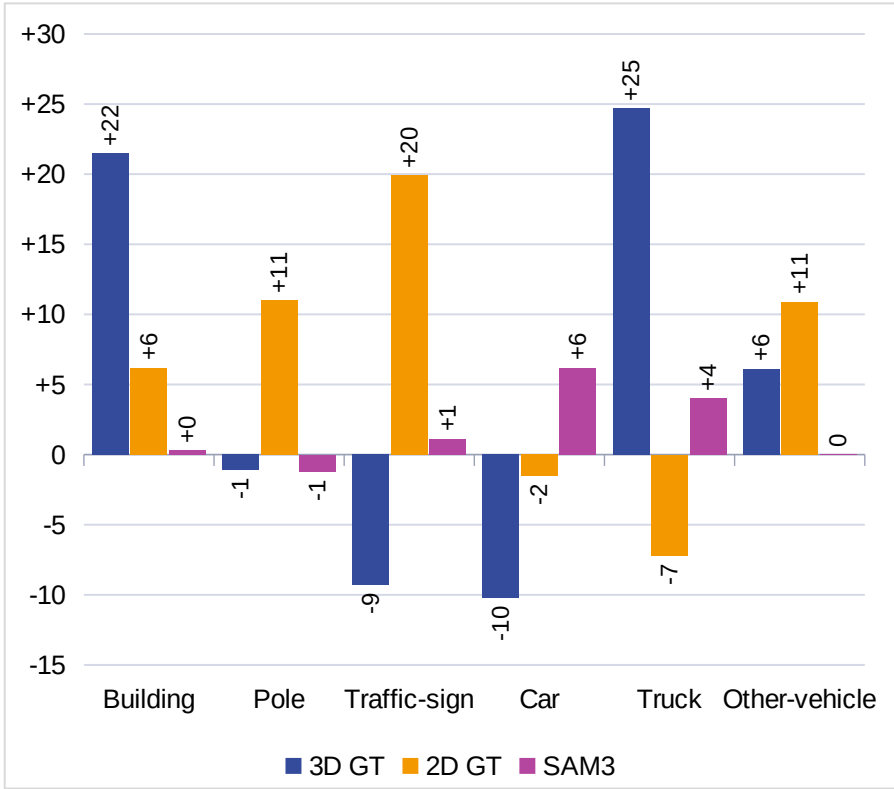
Experimental Results

3-step, final step: per-class IoU where replay matters

IoU without and with replay



Change from replay (× 10^{-3})

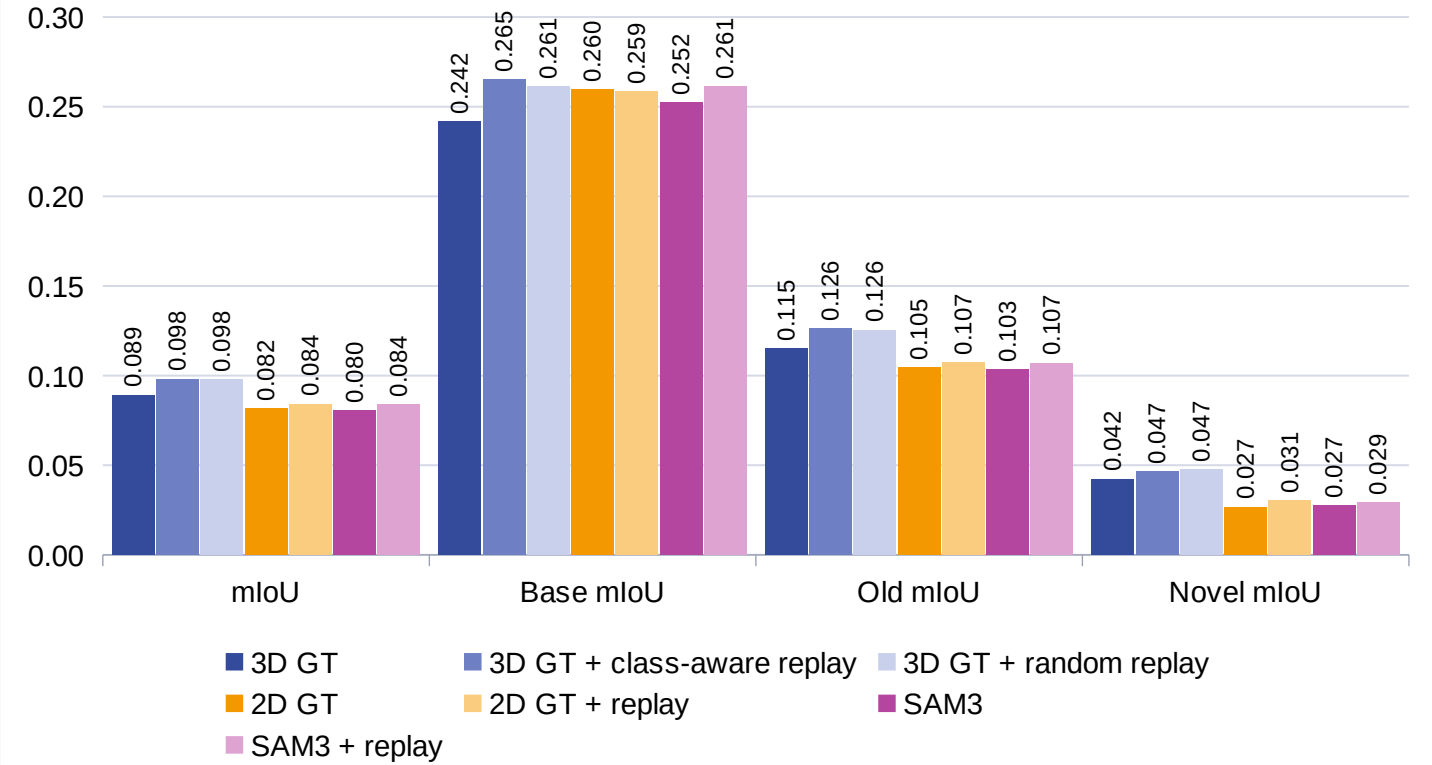


Replay helps 3D GT most on truck (+0.0247) and building (+0.0215), 2D GT on traffic-sign and pole. SAM3 barely changes: 16 of 19 classes move by < 0.005.

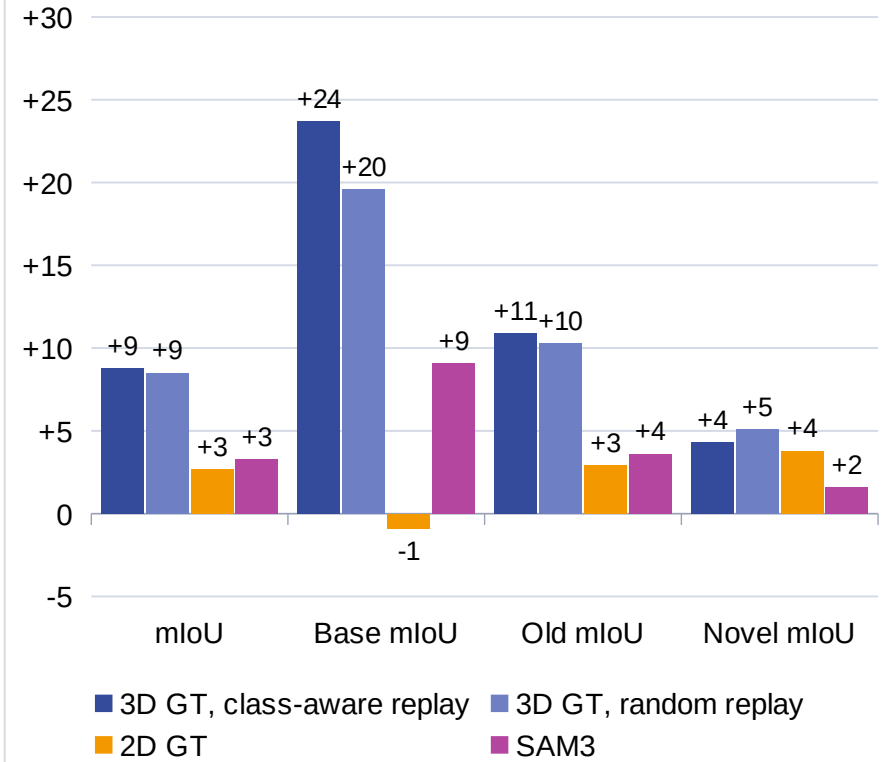
Experimental Results

5-step, final step: what replay adds

Without and with replay (49 frames)



Change from replay (× 10⁻³)

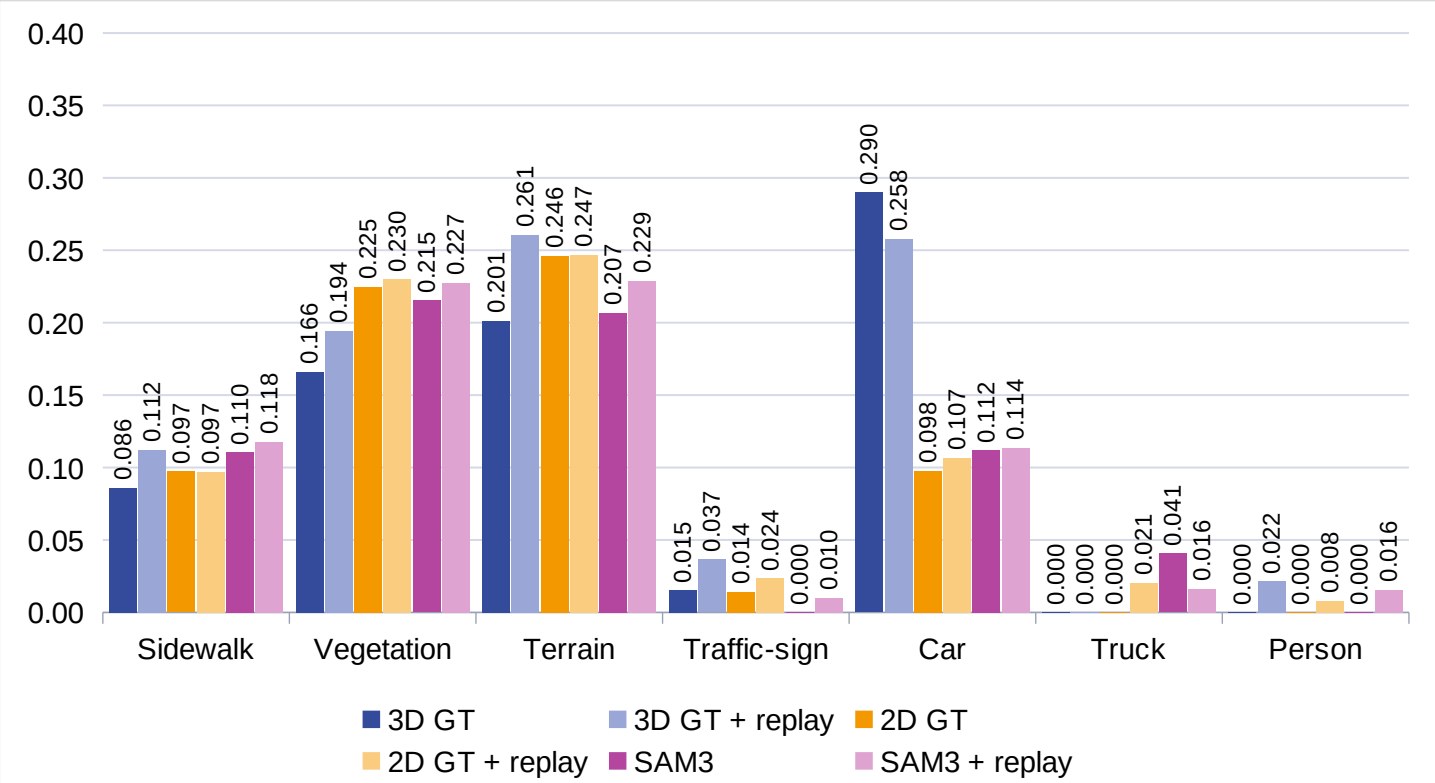


Replay helps most with 3D labels: Old mIoU 0.1152 → 0.1261 (+9.5%); random selection is almost as good (0.1255). 2D GT and SAM3 gain about +0.003 mIoU.

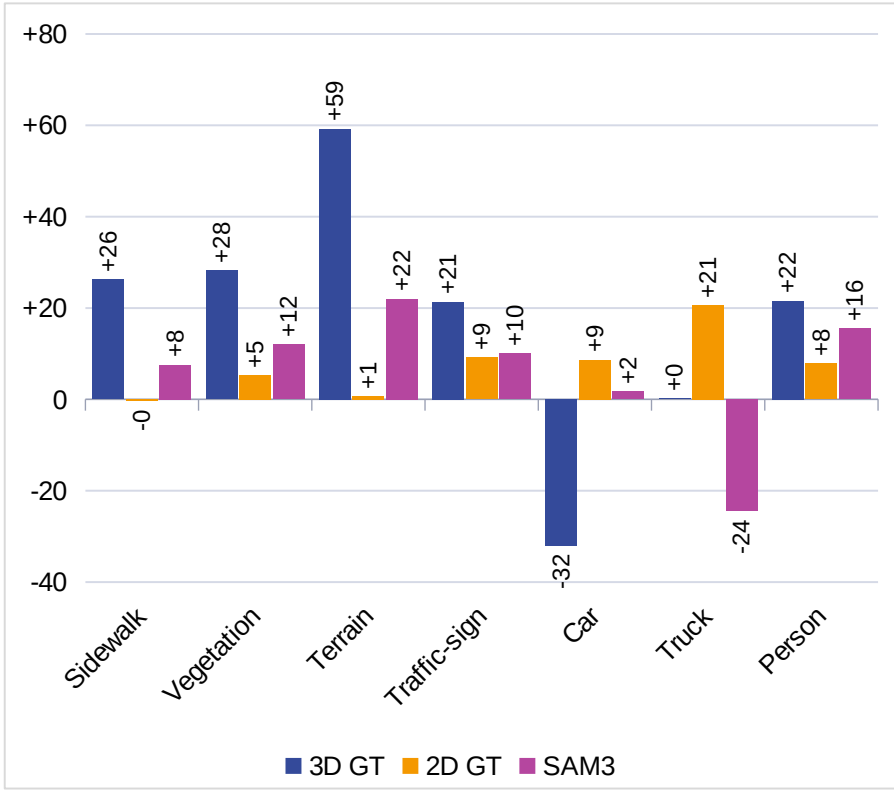
Experimental Results

5-step, final step: per-class IoU where replay matters

IoU without and with replay



Change from replay (× 10⁻³)



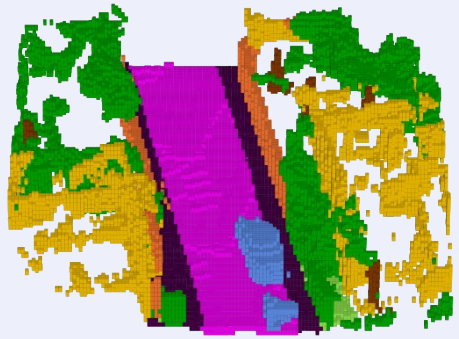
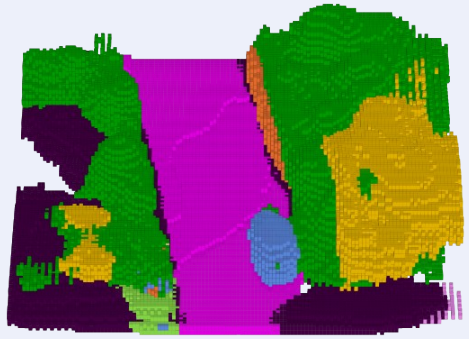
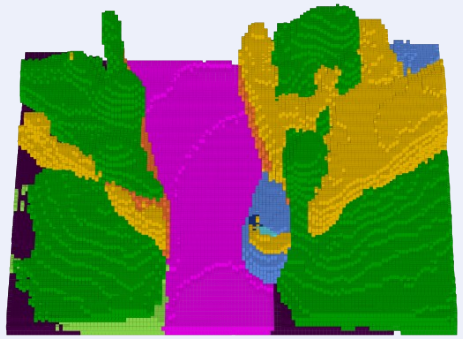
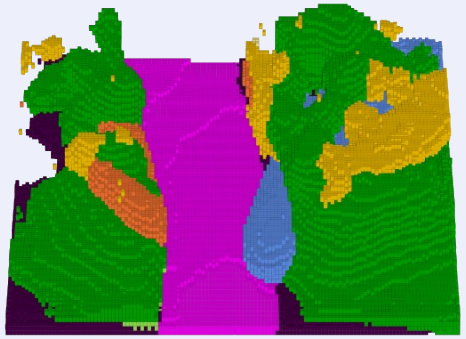
Replay helps 3D GT most on terrain (+0.0592), vegetation (+0.0282) and sidewalk (+0.0262); it brings person back (0 → 0.0216) but costs car (−0.0320). 2D GT gains most on truck (+0.0206), SAM3 on terrain (+0.0220).

Experimental Results

Qualitative results: the street and the car are recovered

Sequence 08



Ground truth	3D GT	2D GT	SAM3
			
LiDAR labels	car IoU 0.258 · road 0.494	car IoU 0.107 · road 0.461	car IoU 0.114 · road 0.472

road · sidewalk · vegetation · terrain · building · car · pole

Road and car are recovered by all three label sources; the car is reconstructed best with 3D GT

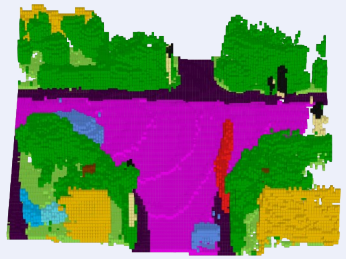
Experimental Results

Qualitative results: what replay changes



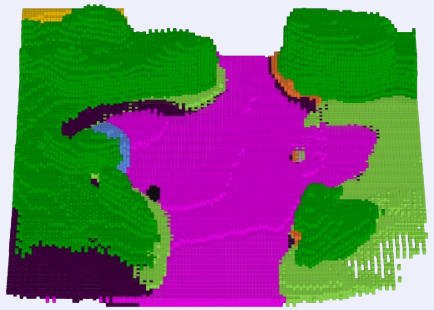
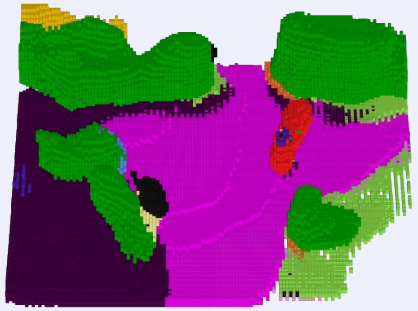
Sequence 08

Ground truth



LiDAR labels

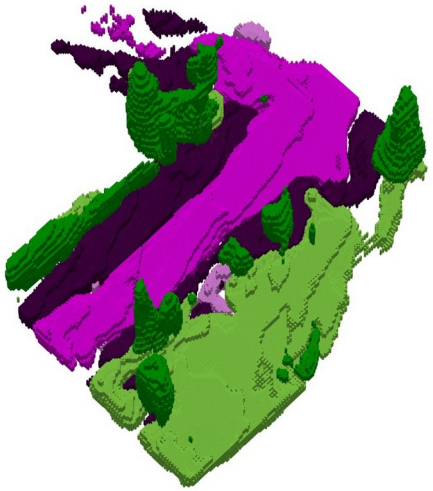
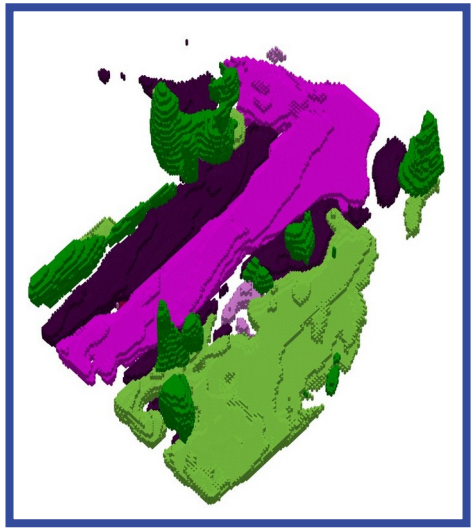
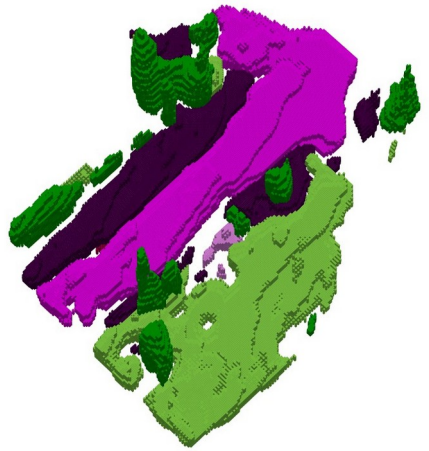
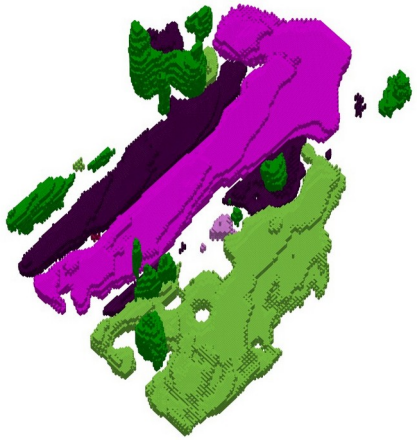
road
sidewalk
vegetation
terrain
building
car
person
traffic-sign

	3D GT	2D GT	SAM3
without replay	 old-class mIoU 0.137	 old-class mIoU 0.116	 old-class mIoU 0.106
with replay	 **old-class mIoU 0.170 (+0.033)**	 **old-class mIoU 0.145 (+0.029)**	 **old-class mIoU 0.135 (+0.029)**

Replay brings back the pedestrian (red) and the traffic sign: both are 0.00 IoU without it, for all three sources

Ablation

Pseudo-label confidence threshold τ

τ = 0.6	τ = 0.7	**τ = 0.8**	τ = 0.9	τ = 0.95
				
noisy	cleaner	**balanced**	fragmenting	sparse

Teacher pseudo-labels, frame 000025 of sequence 08

Conclusion

Answers to the research questions, and next steps

Answers to the research questions

RQ1
New classes

Yes, 2D masks reach 85% (2D GT) and 83% (SAM3) of the 3D-GT mIoU, but on the new classes only 62% and 53% of 3D GT.

RQ2
Replay

Yes, a 49-frame replay buffer helps as we saw in the 5-step setting

RQ3
3 vs. 5 steps

Longer chains forget more, and replay matters more: mIoU went from 0.2526 to 0.0892 after 5 steps vs. 0.2527 to 0.1180 after 3

Next steps

Other replay strategies

Frequency-aware budget
reserve more of the 49 frames for the rarest old classes

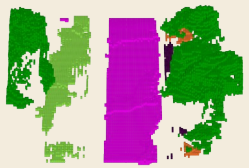
Herding
keep the frames closest to each class mean in feature space

3D foundation models

3D pseudo-labels from 3D foundation models, combined with SAM3

Thank you
Questions and discussion

Old classes: pseudo labels



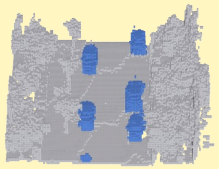
pseudo-labels where the previous model is confident (≥ 0.8)

Old classes: replay buffer



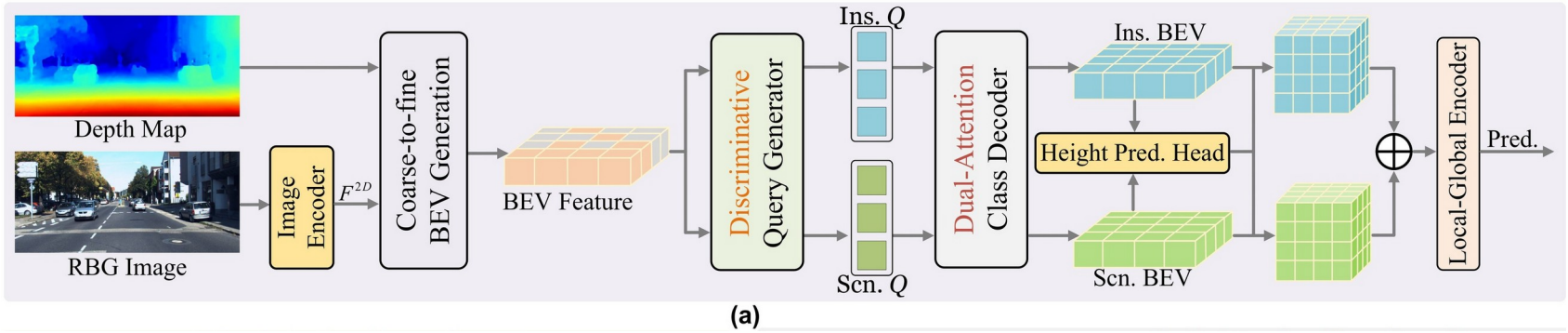
49 stored frames with real old-class labels

New classes: label source



3D GT, 2D GT or SAM3 labels of this step's classes

Student: DISC [1]



output head grows



one node per class

old: copied

new: added

3D losses *or* **2D render CE** Σ **Pseudo-label CE** Σ **Replay CE** → ✓ **Training complete** Becomes the checkpoint of step t+1

[1] Liu et al., "Disentangling Instance and Scene Contexts for 3D Semantic Scene Completion", ICCV 2025

Additional Slides

Metrics at every step, both splits



Additional Slides

3-step split: overall metrics at every step

mIoU

Label source	Step 0	Step 1	Step 2
3D GT	*0.2527*	**0.1738**	0.1180
3D GT + replay	*0.2527*	0.1737	**0.1203**
2D GT	*0.2527*	0.1576	0.1003
2D GT + replay	*0.2527*	0.1587	0.1028
SAM3	*0.2527*	0.1558	0.0977
SAM3 + replay	*0.2527*	0.1536	0.0971

Base mIoU

Label source	Step 0	Step 1	Step 2
3D GT	*0.2527*	**0.2557**	0.2526
3D GT + replay	*0.2527*	0.2542	**0.2544**
2D GT	*0.2527*	0.2519	0.2420
2D GT + replay	*0.2527*	0.2514	0.2423
SAM3	*0.2527*	0.2513	0.2453
SAM3 + replay	*0.2527*	0.2484	0.2439

Old mIoU

Label source	Step 0	Step 1	Step 2
3D GT	–	**0.2557**	0.1696
3D GT + replay	–	0.2542	**0.1726**
2D GT	–	0.2519	0.1528
2D GT + replay	–	0.2514	0.1576
SAM3	–	0.2513	0.1526
SAM3 + replay	–	0.2484	0.1512

Novel mIoU

Label source	Step 0	Step 1	Step 2
3D GT	–	0.0754	0.0559
3D GT + replay	–	**0.0772**	**0.0585**
2D GT	–	0.0445	0.0348
2D GT + replay	–	0.0474	0.0384
SAM3	–	0.0412	0.0295
SAM3 + replay	–	0.0399	0.0293

Bold = best in the column. Old = classes of earlier steps, Novel = classes added at that step; step 0 has neither (–). Replay = 49 stored frames. Step 0: mean of the six per-class IoUs.

Additional Slides

5-step split: overall metrics at every step

mIoU

Label source	Step 0	Step 1	Step 2	Step 3	Step 4
3D GT	*0.2526*	0.1546	0.1328	0.1186	0.0892
3D GT + replay (class-aware)	*0.2526*	**0.1598**	0.1382	0.1232	**0.0980**
3D GT + replay (random)	*0.2526*	0.1593	**0.1409**	**0.1267**	0.0977
2D GT	*0.2526*	0.1563	0.1230	0.1054	0.0815
2D GT + replay	*0.2526*	0.1569	0.1235	0.1065	0.0842
SAM3	*0.2526*	0.1547	0.1216	0.1026	0.0804
SAM3 + replay	*0.2526*	0.1581	0.1260	0.1057	0.0837

Base mIoU

Label source	Step 0	Step 1	Step 2	Step 3	Step 4
3D GT	*0.2526*	0.2485	0.2403	0.2379	0.2416
3D GT + replay (class-aware)	*0.2526*	0.2589	0.2521	0.2580	**0.2653**
3D GT + replay (random)	*0.2526*	0.2580	0.2526	0.2530	0.2612
2D GT	*0.2526*	**0.2644**	0.2598	0.2580	0.2595
2D GT + replay	*0.2526*	0.2618	0.2572	0.2564	0.2586
SAM3	*0.2526*	0.2595	0.2555	0.2519	0.2523
SAM3 + replay	*0.2526*	0.2634	**0.2607**	**0.2592**	0.2614

Old mIoU

Label source	Step 0	Step 1	Step 2	Step 3	Step 4
3D GT	–	0.2485	0.1450	0.1315	0.1152
3D GT + replay (class-aware)	–	0.2589	0.1521	0.1382	**0.1261**
3D GT + replay (random)	–	0.2580	0.1527	**0.1390**	0.1255
2D GT	–	**0.2644**	0.1578	0.1219	0.1045
2D GT + replay	–	0.2618	0.1584	0.1242	0.1074
SAM3	–	0.2595	0.1548	0.1205	0.1034
SAM3 + replay	–	0.2634	**0.1595**	0.1252	0.1070

Novel mIoU

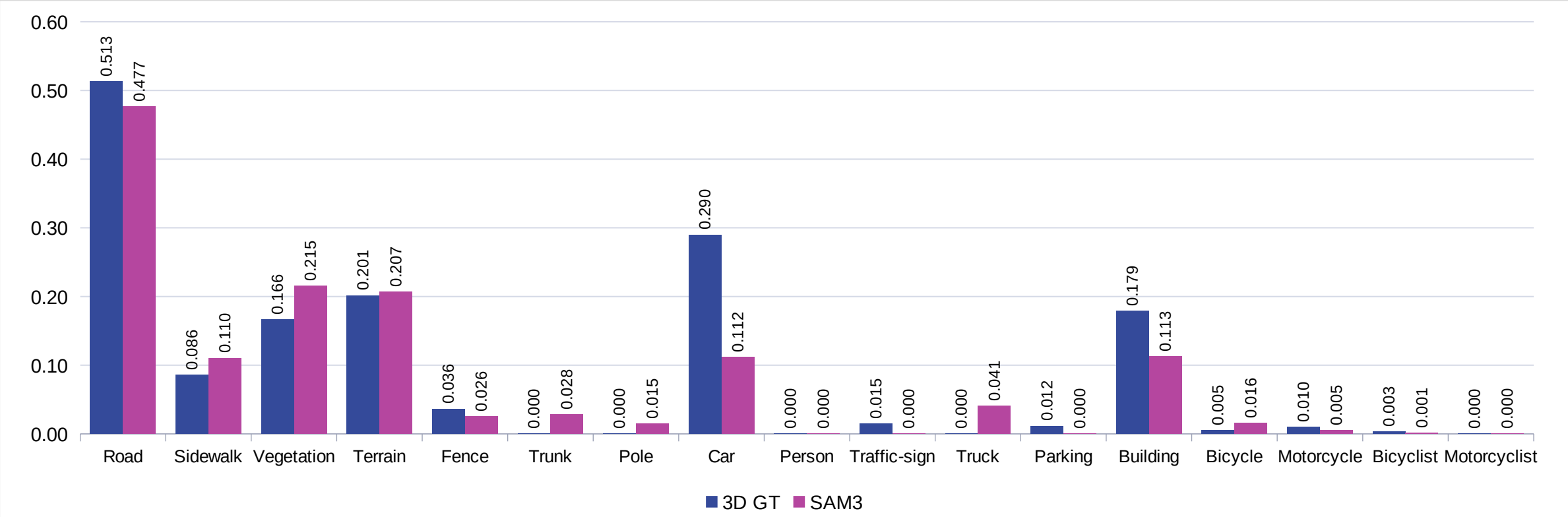
Label source	Step 0	Step 1	Step 2	Step 3	Step 4
3D GT	–	**0.0295**	0.0612	0.0655	0.0423
3D GT + replay (class-aware)	–	0.0277	0.0622	0.0633	0.0466
3D GT + replay (random)	–	0.0278	**0.0665**	**0.0706**	**0.0474**
2D GT	–	0.0122	0.0318	0.0376	0.0267
2D GT + replay	–	0.0170	0.0343	0.0399	0.0305
SAM3	–	0.0150	0.0324	0.0363	0.0274
SAM3 + replay	–	0.0177	0.0362	0.0375	0.0290

Bold = best in the column. Old = classes of earlier steps, Novel = classes added at that step; step 0 has neither (–). Replay = 49 stored frames.

Additional Slides

5-step, final step: per-class IoU, 3D GT vs. SAM3

IoU per class after step 4, no replay

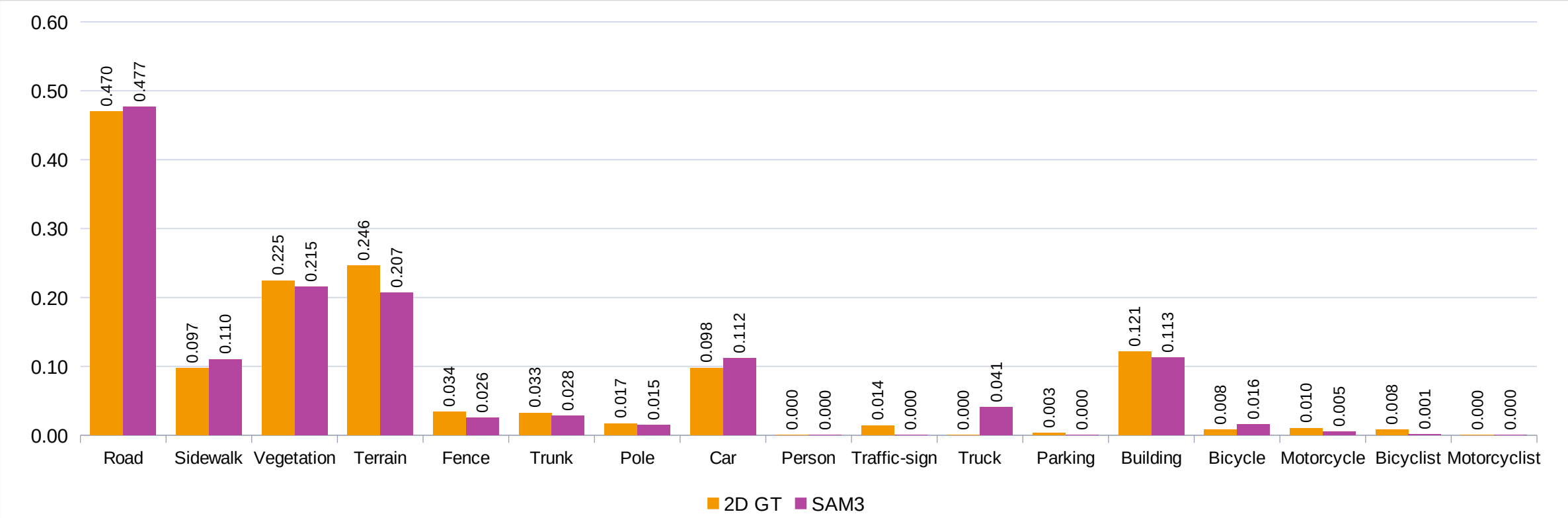


3D GT leads on large classes: car (0.290 vs 0.112), building (0.179 vs 0.113), road (0.513 vs 0.477); SAM3 leads on vegetation, truck, trunk and sidewalk. Both are near zero on person, bicyclist and motorcyclist. mIoU 0.0892 vs 0.0804.

Additional Slides

5-step, final step: per-class IoU, 2D GT vs. SAM3

IoU per class after step 4, no replay

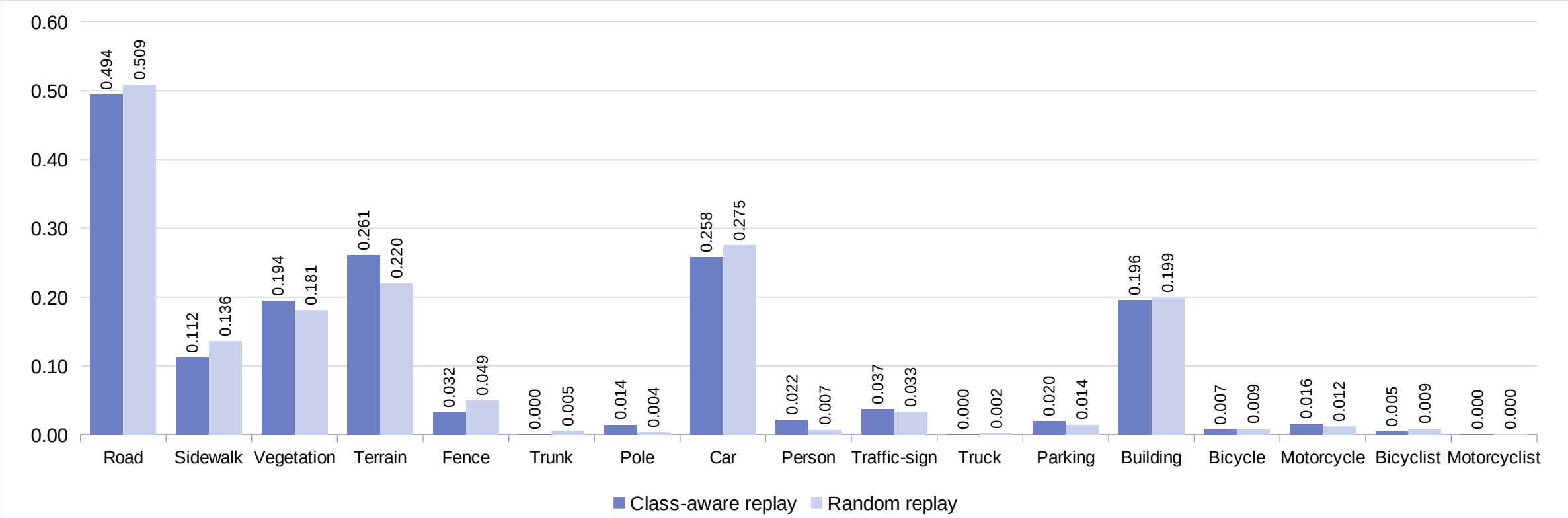


2D GT and SAM3 are within 0.015 of each other on 15 of 17 classes; the exceptions are truck (SAM3 0.041 vs 0.0000) and terrain (2D GT 0.246 vs 0.207). Both trail 3D GT on car and building. mIoU 0.0815 vs 0.0804.

Additional Slides

5-step, final step: per-class IoU, class-aware vs. random replay

IoU per class after step 4, 3D GT with 49 replay frames (classes in the order they are added, step 0 to step 4)



The two strategies are almost tied overall: mIoU 0.0980 (class-aware) vs 0.0977 (random); class-aware is ahead on terrain (0.261 vs 0.220), vegetation and person, random on sidewalk (0.136 vs 0.112), fence, car and road. Differences are within 0.025 on every class except terrain.